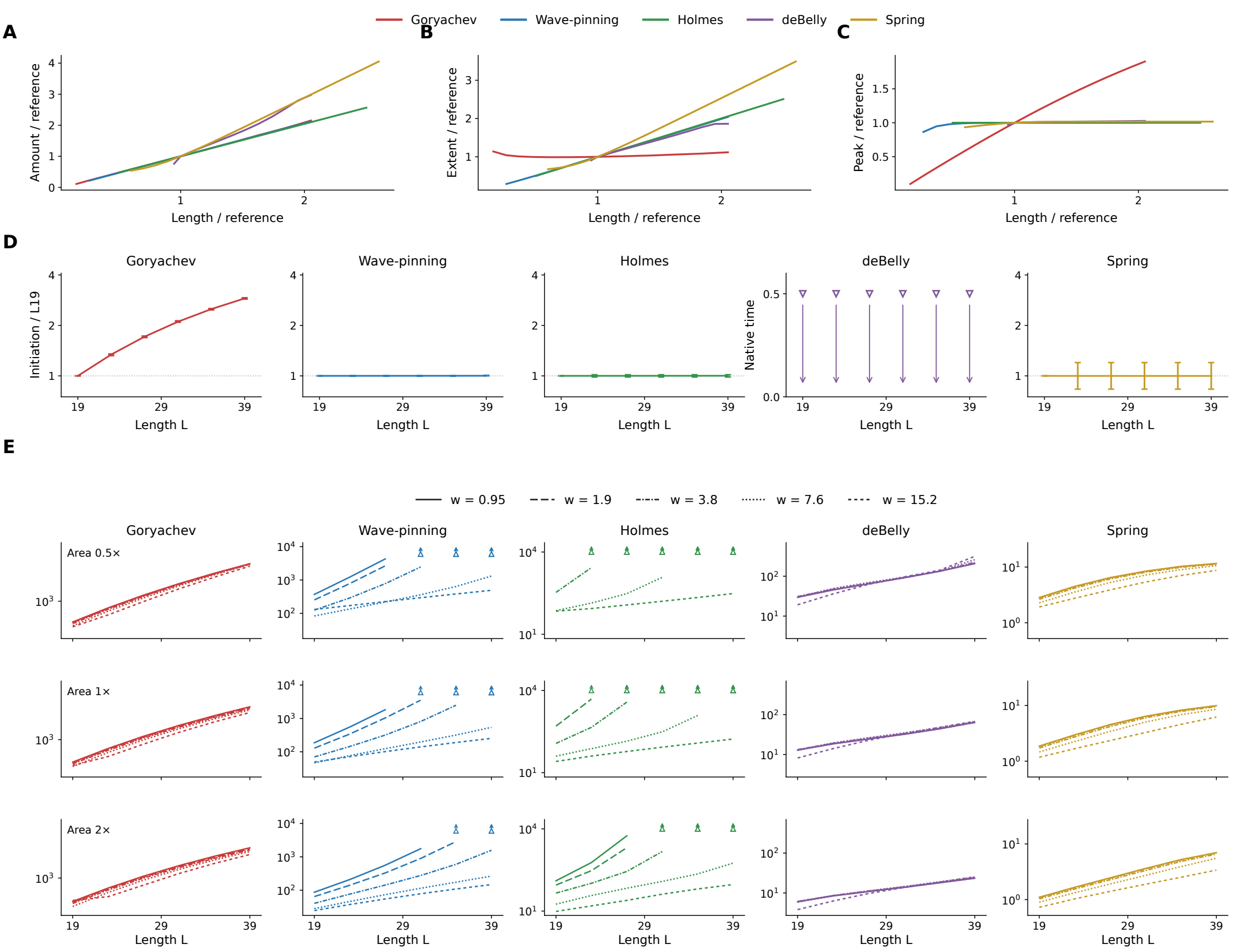

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
B
Extent / reference
Length / reference
C
Peak / reference
Length / reference
D
Goryachev
Wave-pinning
Holmes
deBelly
Spring
Initiation / L19
Native time
Length L
E
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
Length L
F
Pulse-2 gain / headroom
Gap-end / pulse 1
Open circles: below activity gate at x = 0. Size increases from light (L19) to dark (L39).

G
Goryachev
lost
Wave-pinning
kept
Holmes
kept
deBelly
lost then commits
Spring
lost then commits
Challenger share
Time / pulse
L39
H
WGD−
WGD+
Initiation-limited
Goryachev
Saturating activation
Wave-pinning / Holmes
Slow non-activity state
deBelly / Spring
Biosensor
(schematic)
Time (schematic units)
I
Replacement time
(schematic)
narrow
broad
Width at fixed dose

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures

Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

A, Incumbent excess amount normalized to the model reference, across the same 97 retained states.

B, Incumbent extent normalized to the model reference, for those same states.

C, Incumbent peak normalized to the model reference, for those same states.

D, Initiation is the time from cue onset until maximum in-cue activity minus the profile minimum exceeds 10% of the model's L19 incumbent excess peak. Area 1, width 3.8; times are normalized to L19. All six deBelly upper bounds are 0.5 native units (downward arrows); first-update sampling leaves its size dependence unresolved.

E, Replacement time (first eligible challenger share ≥ 0.75) versus length, with a line per width and rows for areas 0.5, 1 and 2. Upward triangles/arrows are lower bounds: the cue never replaced within the horizon (6,000 Wave-pinning; 10,000 Goryachev/Holmes; 2,000 deBelly; 100 Spring, native units). Broad cues rescue Holmes across all lengths; narrow-cue censoring does not establish a growing penalty. Spring native widths equal the displayed widths divided by 38.

F, Gap-1 activity kept versus pulse-2 gain as a fraction of headroom (Eq. 1). Four equal pulses, total ON 1.5 times the model-length commitment calibration; gaps 0–16 pulse durations. Holmes L31–39 uses the L27 calibration. Other models are grey; lightness indicates L. Open circles at x = 0 code activity below the existing visibility gate. At the left wall, deBelly/Spring gain on pulse 2 despite no visible activity; Goryachev never does. The deBelly L19/gap16 first-gap endpoint was checked against native fields.

G, L39 gap4 challenger-share histories, labelled kept / lost / lost then commits; shading marks cue ON. deBelly includes full recovery; others show one post-train pulse duration. Continuous raw shares include below-gate values. deBelly's later timing and fields remain numerically sensitive (supplement).

H, Prospective biosensor histories for the same pulse train in WGD− and enlarged WGD+ cells. Initiation-limited resetting (Goryachev), saturating activation with gap retention (Wave-pinning/Holmes), and progress retained after activity loss (deBelly/Spring) give distinct dimensionless schematic histories.

I, Prospective equal-dose width responses for those classes: weak width rescue, strong width rescue, and weaker width rescue. Solid grey: WGD−; dashed black: WGD+. These are qualitative assay predictions.

$$x = \frac{s_{\mathrm{gap}\,1}}{s_1}, \qquad y = \frac{s_2 - s_1}{1 - s_1} \tag{1}$$

s_1 and s_2 are challenger shares at pulse ends 1 and 2; s_gap1 is the first-gap-end share. An observed endpoint below the existing visibility gate is coded as zero visible share in F; its raw

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

value remains in the CSV.